1 Motivation and assumptions

Why do we need both thermodynamics and statistical mechanics?

Thermodynamics allows one to relate macroscopic quantities without performing microscopic calculations. It provides:

- the *relation* between macroscopic quantities (e.g. the pressure, p, volume, V, and temperature T of a gas in the equation of state), or deducing the work performed in an adiabatic compression in terms of initial and final macroscopic variables. If one measures some quantities one can deduce others and as such is useful both in considering refrigeration, heating, compression of gases and their applications;
- the construction of the analogue to potential energy (eg free energy from the internal energy) to determine equilibrium when minimised reflecting the variables which are controlled (volume or pressure etc);
- the capacity to relate something which is difficult to measure (e.g. anything involving entropy, S) to a readily measurable quantity using one or more Maxwell relations. For example:
$$\left(\frac{\partial S}{\partial V}\right)_T = \left(\frac{\partial p}{\partial T}\right)_V .$$

Thermodynamics does *not* allow the calculation of the internal energy, $U(S, V, N)$ (where N is the number of particles), and the quantities one may deduce from that: a specific heat, C_V, compressibility, $\kappa = (\partial V/\partial p)_S$, etc and the entropy. The other thermodynamic potentials are also not deducible: the enthalpy, $H(S, p, N)$, the Helmholtz free energy, $F(T, V, N)$ and the Gibbs free energy, $G(T, p, N)$. One may measure the specific heat and, by integration, deduce the internal energy, but not predict the result or "understand" it.

Statistical mechanics allows the calculation (in principle!) of, for example, the internal energy of a gas (and the other thermodynamic potentials) from knowledge of the masses and numbers of moles of the constituent molecules and atoms (and any interactions if not an ideal gas). It provides a link with classical or quantum mechanics.

Both approaches are useful depending on the contexts.

1.1 Simplicity with large numbers

Thermodynamics deals with macroscopic (implicitly averaged) quantities. For example the pressure in a balloon, the temperature of a body etc. In the case of the balloon, the pressure (either inside or outside) may be thought of microscopically as the result of molecules being reflected from its surface, with their momenta perpendicular to the surface, approximately, reversed after the collision. (The tension of the rubber membrane of the balloon balances the pressure difference between inside and outside.)

But one does not see a balloon quivering – so the deviations from average pressure must be proportionately very small. It is plausible this is because of the very large number of molecules in the gas and each impulse upon reflection being very small. The lack of quivering implies the probability distribution of the pressure in any part of the surface must be very strongly peaked around the average value. We will quantify this, in part, presently.

A simple example of such a peak in a probability distribution when large numbers are involved is in a large number, N, of coin tosses. Let $W(N,n)$ be the number of sequences of tosses yielding n heads:

$$W(N,n) = \frac{N!}{(N-n)!n!} \ .$$

This may be turned into a probability of observing n, $p(N,n)$, by normalising $W(N,n)$:

$$p(N,n) = \frac{W(N,n)}{\sum_{n=0}^{N} W(N,n)} = 2^{-N} W(N,n) \ .$$

(Check this is normalised.) The factor of 2^{-N} comes from noting:

$$2^N = (1+1)^N = \sum_{n=0}^{N} 1^n 1^{N-n} \ W(N,n) = \sum_{n=0}^{N} W(N,n) \ ,$$

i.e. the definition of binomial coefficients.

If we pick a particular realisation, i.e. string of $h, t, h, t, t, t \cdots$, then the probability of that being less than 1% away from the plausible most likely case of half heads, half tails ($n = N/2$) is shown in table below.

The implication is the probability of seeing a result which differs significantly from $n = N/2$ is vanishingly small as $N \to \infty$: the probability distribution becomes very narrow. One almost always sees the mean behaviour – like the pressure in the balloon. In terms of the function $W(N,n)$, it is very sharply peaked at $n = N/2$ – i.e. the number of ways to have an equal number of heads and tails represents a sharp maximum in W.

Number tosses, N	Percentage of heads being within 1 percent $N/2$
10^3	0.248
10^4	0.683
10^5	0.998
10^6	$1 - 1.5 \times 10^{-23}$
10^8	$1 - 2.7 \times 10^{-2174}$

Table 1: Data from Baierlein.

2 Energetically isolated systems

In this section we describe isolated systems which allow no heat exchange with the external world – so energy is conserved. E will denote the total energy of the system. Firstly we will see how to denote the state of the system, at a microscopic level, economically. For a classical *isolated* system there is no friction[1] at a microscopic level. Thus the equations of motion take a very simple form: Newton's second law, "$F = ma$", for each particle. To solve these differential equations (in principle) one needs to know the positions and velocities of each particle at some time, say t_0, to provide the initial conditions. Because the momentum is a more natural variable than the velocity, it is used instead.

2.1 Phase space – classical statistical mechanics

Phase space is the arena for statistical mechanics – it allows one to represent in *one place* the dynamics of all of the constituent atoms/molecules. instead of specifying N position vectors and momenta, one point is specified.

For a monatomic classical gas of N atoms of mass m, we denote (for reasons buried in history) the position of the nth atom at time t by $\mathbf{q}_n(t)$ and the atom's momentum by $\mathbf{p}_n(t)$.

In three dimensions, the set of the $3N$ coordinates, $\{\mathbf{q}_n(t)\}$, and $3N$ components of the momenta, $\{\mathbf{p}_n(t)\}$, together determine a point in the $6N$-dimensional *phase space*. This represents a complete specification of the gas at a given instant, t. (And from that, with the equations of motion, one may determine the state for any subsequent time.)

Since the system is classical the equations of motion for the coordinates and momenta are determined by Newton's second law which may be represented by:

$$m\dot{\mathbf{q}}_n = \mathbf{p}_n \quad \text{and} \quad \dot{\mathbf{p}}_n = -\frac{\partial}{\partial \mathbf{q}_n} V\left(\mathbf{q}_1, \mathbf{q}_2, \cdots, \mathbf{q}_N\right) \ , \tag{2.1}$$

[1] Being careful we would say uncharged particles so no issue of the electromagnetic field.

where V is the total potential energy which is often decomposed into

$$V\left(\mathbf{q}_1, \mathbf{q}_2, \cdots, \mathbf{q}_N\right) = \sum_{n=1}^{N} u_1\left(\mathbf{q}_n\right) + \tfrac{1}{2} \sum_{\substack{n,n'=1,\\ n\neq n'}}^{N} u_2\left(\mathbf{q}_n - \mathbf{q}_{n'}\right) \ ,$$

with $u_1\left(\mathbf{q}_n\right)$ representing, for example, each atom interacting with the "container". The potential involving the $u_2\left(\mathbf{q}_n - \mathbf{q}_{n'}\right)$ describes any inter-atomic interactions – which would be zero for an ideal gas. (We do not consider velocity-dependent forces such as the Lorentz force. There is no friction – we are describing the system microscopically.)

If one replaces $\mathbf{p}_n$ in the second equation in Eq. (2.1) using the first equation one regains the familiar form "$m\mathbf{a} = m\ddot{\mathbf{q}} = \mathbf{F}$". In the form of Eq. (2.1) they are called *Hamilton's equations*.

The energy, or Hamiltonian $\mathcal{H}$, is

$$\begin{aligned}\mathcal{H}\left(\mathbf{q}_1, \mathbf{q}_2, \cdots, \mathbf{q}_N; \mathbf{p}_1, \mathbf{p}_2, \cdots, \mathbf{p}_N\right) &= \underbrace{\sum_{n=1}^{N} \frac{\mathbf{p}_n^2}{2m}}_{\text{kinetic energy}} + \underbrace{V\left(\mathbf{q}_1, \mathbf{q}_2, \cdots, \mathbf{q}_N\right)}_{\text{potential energy}} \\ &= \mathcal{H}(Q, P) \ , \end{aligned} \tag{2.2}$$

where we have introduced the symbolic shorthand: Q denotes all the components of the coordinates and P all components of the momenta.

2.1.1 Geometry of the motion

Firstly consider a sphere in 3 dimensions with radius a, with coordinates x, y and z. The equation of the sphere's surface is

$$a^2 = x^2 + y^2 + z^2 \ .$$

The surface is two dimensional – you only need two polar angles to identify a point on it, but the space it is in is 3d. From an algebraic point of view one has one equation which reduces the number of variables from three to two – you could eliminate z, for example, via

$$z = \pm\sqrt{a^2 - (x^2 + y^2)} \ .$$

For phase space, the dimension is $6N$, for N particles in three dimensions – three components of each of the position vector and the momentum for each particle.

Energy is conserved so we have one equation

$$\begin{aligned} E &= \mathcal{H}(\mathbf{q}_1, \cdots, \mathbf{q}_N; \mathbf{p}_1, \cdots, \mathbf{p}_N) \\ &= \sum_{n=1}^{N} \frac{\mathbf{p}_n^2}{2m} + V(\mathbf{q}_1, \cdots, \mathbf{q}_N) \ , \end{aligned}$$

which could be used to eliminate one of the variables, say the x-component of $\mathbf{q}_1$, the position vector of the first particle. There would then be $6N-1$ variables left free. So the surface of constant energy is a $6N-1$-dimensional space, within the full $6N$-dimensional phase space.

For an ideal gas – with no interactions between the particles – we may be more explicit. The position vectors are unconstrained except by the walls of the "container", to reside in a volume V. However the momentum variables are more interesting. The energy only involves them:

$$E = \sum_{n=1}^{N} \frac{\mathbf{p}_n^2}{2m}$$
$$\Rightarrow \quad 2mE = a^2 = \sum_{n=1}^{N} \mathbf{p}_n^2 \ .$$

This is the equation of a sphere, of radius $\sqrt{2mE}$, in the $3N$-dimensional space of the momentum variables (remember the factor of 3 in $3N$ comes from the three components of each $\mathbf{p}_n$). So the full "surface" is a product of N cubes, each with volume V (if the container is cubic), times a $3N-1$-dimensional surface, which is the surface of a "hypersphere" in $3N$-dimensions.

If there are more conserved quantities than the energy (eg angular momentum, or the centre of mass momentum), this allows more variables to be eliminated, so the dimension of the surface drops. If all the momenta were conserved, then the surface becomes $3N$-dimensional.

2.2 Equilibrium and Poincaré recurrence (nonexaminable)

Consider gas in a thermally insulated container initially divided into two regions: one region has all the atoms within it, the other region is empty. Remove the partition. The gas spreads out and experimentally eventually settles down to a state where the macroscopic variables such as pressure are uniform over the gas. This is *equilibrium* , which we will focus on in this module – not the "settling down". *Non-equilibrium* statistical mechanics is still very much under development, despite the progress that Boltzmann and others have made over the last 150 years.[2]

One paradox (the *Loschmidt paradox*) is that the equations of motion, Eq. (2.1), are *reversible*: if we reverse all the momenta at time $t = \tau$ and wait for a time τ, the atoms will be back where they were initially, at $t = 0$ (but with their initial velocities reversed). So in the situation of the last paragraph, there is a solution to the equations of motion where all the atoms rush back to one side of the total volume. Why is that never observed although it is an allowed time-evolution?

[2] It has been discovered there are some systems which **never** settle down. See four minute talk by Dr Curt von Keyserlingk on *time crystals*.

One might have an intuitive notion that this would be very unlikely (correctly). But, surprisingly, if we waited *long enough* we would see the particles back at their initial positions and momenta, a *Poincaré recurrence*, *with a certain accuracy* ,

$$
\begin{aligned}
& |\mathbf{q}_n(t=0) - \mathbf{q}_n(t=\tau)| < a_q \epsilon \\
& |\mathbf{p}_n(t=0) - \mathbf{p}_n(t=\tau)| < a_p \epsilon \\
\text{(Introduce scaled dimensionless variables)}\ & |\tilde{\mathbf{q}}_n(t=0) - \tilde{\mathbf{q}}_n(t=\tau)| < \epsilon \\
& |\tilde{\mathbf{p}}_n(t=0) - \tilde{\mathbf{p}}_n(t=\tau)| < \epsilon
\end{aligned}
$$

for all n. Here, a_q has the dimension of length and a_p the dimension of momentum, with the tilde variables thus being dimensionless.

Then we might anticipate that the smaller we make ϵ (I.e the closer to the starting point in phase space), the longer we must wait.

To see there must be a recurrence, imagine the point in phase space moving with a speed c in phase space the and we construct a "hyper-cylinder" around its trajectory with "radius" ϵ, providing a "worm-like" trajectory (WLT). Using the scaled dimensionless variables, the phase space volume swept out is roughly (ignoring "π-like" factors and being very sloppy about momentum- and position-like directions),

$$\Omega_{\text{s}} = ct(\epsilon)^{6N-1} \tag{2.3}$$

The recurrence will occur when this "worm-like" trajectory intersects itself. Since the volume in phase-space is bounded, this must happen – it cannot avoid itself forever. In the spatial coordinates the boundary is set by the volume V^N, in the momentum variables by "volume" (really a hyper-area), S_n, of the locus of the momentum hypersphere

$$E_{\text{tot}} = \frac{a_p^2}{2m} \sum_{n=1}^{N} (\tilde{\mathbf{p}})^2 \ .$$

with radius

$$R = \sqrt{\frac{2m\, E_{\text{tot}}}{a_p^2}} \ ,$$

where

$$S_n \underset{N\to\infty}{\sim} R^{3N-1}([3N-1]/2)^{-[3N-1]/2}\,(\mathrm{e}\pi)^{(3N-1)/2} \ .$$

Note that in a sensible limit, the total energy is proportional to N, say $E_{\text{tot}} = NE_0$. Upon substitution we find for the total phase space volume:

$$\Omega_{\text{tot}} \underset{N\to\infty}{\sim} \left(\frac{2\pi\mathrm{e}}{3}\,\frac{2mE_0}{a_p^2}\right)^{(3N-1)/2} V^N \ .$$

It is very hard to make a precise estimate and the following is a woolly estimate of an upper bound in terms of the time.

To estimate the recurrence time we ask for the time, $t = t_{\text{recur}}$, when

$$\Omega_{\text{tot}} = \Omega_{\text{s}}(t_{\text{recur}})$$

$$\Rightarrow \quad V^N \left(\frac{2\pi \mathrm{e}}{3}\frac{2mE_0}{a_p^2}\right)^{(3N-1)/2} = ct_{\text{recur}}\epsilon^{6N-1}$$

$$\Rightarrow \quad t_{\text{recur}} = \left(\frac{2\pi \mathrm{e}}{3}\frac{2mE_0}{a_p^2}\right)^{(3N-1)/2} V^N \frac{1}{c\epsilon^{6N-1}} \ .$$

So if we have Avogadro's number of atoms, $N \sim 10^{23}$, as $\epsilon \to 0$, i.e. increased accuracy of recurrence, $t_{\text{recur}} \to \infty$ **very** rapidly:

$$t_{\text{recur}} \propto \epsilon^{-6\times 10^{23}} \ .$$

It is difficult to estimate accurately, but the estimates are of the above order or longer.

An insight into the influence of high dimensions comes from a discrete analogue of the motion described above. Imagine that the allowed positions and momenta are not continuous numbers but are discrete, with the allowed values constituting a "lattice". This is immediately clear for the position variables – then the discrete values are like the atomic positions of a crystal lattice. We assume the discrete set forms a cubic lattice in $6N$-dimensional space

The argument above provides an upper bound to a recurrence. One might wonder about "short loops" in the trajectory. Here being in a high dimensional space makes loops harder to form. We illustrate that by considering a random walk on a (hyper)cubic lattice as a discrete version of the dynamics.

We investigate the probability that the trajectory forms a loop of the smallest extent, i.e. 4 lines forming the sides of a square, see Fig. (2.1). Let us envisage the process in two dimensions and then calculate the probability of a short loop in dimension D.

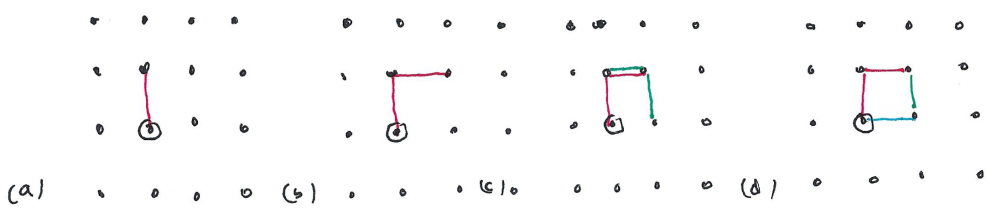

Figure 2.1: A random walk on a square lattice. The circled dot is the origin or starting point the first and subsequent "steps" are in red. (a) The first step may be in any of the four directions from the origin. (b) The second step could return to the origin, or take one of the other three directions. (c) If the latter then we are now forced (to achieve a loop) to take one of the first two steps in the opposite direction and then the other step in the opposite direction.

The total number of "walks" of four steps are 4^4. The number of completely retraced paths are 4 involving two sites both revisited twice, 4×3 walks which visit three sites (and enclose

no area) and 4×2 walks which trace out a square (the factor of 2 is from the sense in which it is traversed – clockwise or anticlockwise). I.e. 24 closed walks out of $256 = 4^4$.

For D dimensions, those three categories yield: $2D$, $2D(2D-1)$ and $2D(2D-2)$ walks compared to $(2D)^4$ total number of four step walks. The fraction of the closed walks (or "loops") for $D \to \infty$ is approximately $3/(2D)^2 \to 0$. So loops are very rare in high dimensions.

2.3 Experiments

A real experiment on a sample (of gas for example) usually measures macroscopic quantities such as pressure and volume over some duration of time.

To get a quantitative feel for time scales, say the experiment is on a sample of *ideal*[3] gas of 10 cm linear dimensions, so volume $10^{-3}\,\mathrm{m}^3$. At room temperature the molecules in air are moving at around $500\,\mathrm{ms}^{-1}$, so covering the sample dimensions in a time

$$\tau = \frac{10^{-1}}{500}\,\mathrm{s} = 2 \times 10^{-4}\,\mathrm{s}\ .$$

Perhaps the experiment takes a time $\tau_{\mathrm{exp}} = 10$ minutes, so the ratio of times is

$$\frac{\tau_{\mathrm{exp}}}{\tau} = \frac{600}{2 \times 10^{-4}} = 3 \times 10^6\ ,$$

i.e. the gas atom traverses the container more than a million times during the experiment. And there are around 3×10^{22} molecules contributing to the pressure, ie more than 10^{20} collisions with one wall per second.

Thus if the experiment measures a macroscopic variable, such as pressure, it is an *average* pressure – the individual impacts of the atoms are not resolved, as we observed with the example of the balloon earlier.

The problem with this observation is while the average is measurable, the average is *incalculable* except as an order of magnitude – as we cannot solve the equations of motion for that number of particles.

[3] In reality it is the molecules within a mean free path, λ, of the surface of the container which travel in straight lines. However the result is the same. The atoms travel the distance to the surface in a time $\Delta\tau = \lambda/v$, where v is typical velocity normal to surface. The number of molecules in the layer of thickness λ is $N_\ell = \lambda L^2 n_0$, where L^2 area of surface and n_0 is number density of molecules. Then the number of collisions per unit time is: $N_\ell/\Delta\tau = v n_0 L^2$, independent of λ, and the same as the ideal case.

2.4 The ergodic hypothesis exemplified by ideal gas

The ergodic hypothesis states that the (observed but incalculable) time average, $\overline{f}$, of a function of the coordinates and momenta, $f(Q,P)$ is equal to a (calculable) average, with an appropriate probability distribution, $D(Q,P)$, over phase incorporating the energy being constant, i.e. the "hypersurface" obeying $E = \mathcal{H}(Q,P)$. I.e.

$$\overline{f} = \lim_{T\to\infty} \frac{1}{T}\int_0^T \mathrm{d}t\ f(Q(t),P(t)) = \int \mathrm{d}\Gamma\ f(Q,P)D(Q,P) \tag{2.4}$$

$$= \langle f \rangle\ . \tag{2.5}$$

We impose residing on the surface of constant energy using a delta function, $\delta[E-\mathcal{H}(Q,P)]$. (Remember a delta function is zero except where its argument vanishes; in this case $E - \mathcal{H}(Q,P) = 0$.) How do we turn this into a probability distribution, $D(Q,P)$? Notice the following is normalised (we use the shorthand $\mathrm{d}Q\mathrm{d}P = \prod_{n=1}^{N} \mathrm{d}\mathbf{q}_n \mathrm{d}\mathbf{p}_n$)

$$\begin{aligned} D(Q,P) &= \frac{\delta[E-\mathcal{H}(Q,P)]}{\int \mathrm{d}Q\mathrm{d}P\delta[E-\mathcal{H}(Q,P)]} \\ \Rightarrow \quad \int \mathrm{d}Q\mathrm{d}P\ D(Q,P) &= \int \mathrm{d}Q\mathrm{d}P\ \frac{\delta[E-\mathcal{H}(Q,P)]}{\int \mathrm{d}Q\mathrm{d}P\delta[E-\mathcal{H}(Q,P)]} \\ &= \frac{\int \mathrm{d}Q\mathrm{d}P\delta[E-\mathcal{H}(Q,P)]}{\int \mathrm{d}Q\mathrm{d}P\delta[E-\mathcal{H}(Q,P)]} \\ &= 1\ . \end{aligned}$$

An interpretation of the denominator in the expression for $D(Q,P)$, which provides normalisation, is that it counts every point in phase space where the total energy is of the allowed value, E. Thus we can say, by analogy with the case of coin tosses, that the "number" of configurations, $W(E,V,N)$, (actually it is the density of states per unit energy) is (further compressing the notation, $\mathrm{d}Q\mathrm{d}P = \mathrm{d}\Gamma$)

$$W(N,V,E) = \int \mathrm{d}Q\mathrm{d}P\ \delta[E-\mathcal{H}(Q,P)] = \int \mathrm{d}\Gamma\ \delta[E-\mathcal{H}(Q,P)]\ .$$

The dimensions of $W(E,V,N)$ are not those of a pure number – we will return to that issue soon.

So substituting this into the ergodic hypothesis,

$$\overline{f} = \lim_{T\to\infty} \frac{1}{T}\int_0^T \mathrm{d}t\ f(Q(t),P(t)) = \int \mathrm{d}\Gamma\ f(Q,P)\frac{\delta[E-\mathcal{H}(Q,P)]}{W(E)} \tag{2.6}$$

$$= \int \mathrm{d}\Gamma\ f(Q,P)D(Q,P) \tag{2.7}$$

$$= \langle f \rangle\ . \tag{2.8}$$

This is the simplest assumption – in the sense that all regions of the "hypersurface" of constant energy (its dimension is $6N-1$) are weighted by the probability distribution $D(Q,P)$ **equally**, i.e. not $D(Q,P) \propto \Delta(Q,P)\delta[E-\mathcal{H}(Q,P)]$, so $\Delta(Q,P) = 1$. This is called **"equal a priori probabilities"**.

Geometrically, $W(N, V, E)$ is the area of the hypersurface of constant energy, E, and is plausibly proportional to the **number of states in what we will term the "hyperarea"**. (There is a little subtlety to this terminology which we can ignore.)

2.4.1 Evaluation of hyperarea

Firstly let us consider a single particle. Say it is contained in a volume V and has energy

$$E = \frac{1}{2m}(p_x^2 + p_y^2 + p_z^2) = \frac{p^2}{2m} \ .$$

We may determine the number of states in two equivalent ways. In both it helps to recognise that behind our classical analysis, there is quantum mechanics. The uncertainty principle implies that a single quantum state occupies a region (we ignore 2π's etc, which actually leads to the correct answer[4])

$$\delta q \delta p \simeq \mathsf{h}$$

in phase space.

First method: So to *count* the states we should divide the integral over the 3 coordinates and three components of momentum by h^3.

Now let (as implied above)

$$|\mathbf{p}| = p = \sqrt{2m\,E} \ , \tag{2.9}$$

be the corresponding "radial coordinate" in the momentum variables.

Then the total number of states in the spherical shell between p and $p + \mathrm{d}p$ in combination with the volume V is is:

$$W(p, V)\mathrm{d}p = \frac{V 4\pi p^2 \mathrm{d}p}{\mathsf{h}^3} \ .$$

where $4\pi p^2 \mathrm{d}p$ is the volume of the spherical shell thickness $\mathrm{d}p$. $W(p, V)$ is the number of states per unit momentum.

We originally wanted $W(E, V)$ (the number of states per unit energy), so we may now change variables from p to E, using Eq. (2.9):

$$p = \sqrt{2m\,E} \quad \text{and} \quad \mathrm{d}p = \frac{\mathrm{d}p}{\mathrm{d}E}\mathrm{d}E = \frac{1}{2}\sqrt{\frac{2m}{E}}\mathrm{d}E \ .$$

Thus we see

$$W(p, V)\frac{\mathrm{d}p}{\mathrm{d}E}\mathrm{d}E = W(E, V)\mathrm{d}E = \underbrace{4\pi V \left(\frac{2mE}{\mathsf{h}^2}\right)^{3/2}}_{\text{dimensionless}} \frac{\mathrm{d}E}{2E} \ . \tag{2.10}$$

[4] **Non-examinable**: In fact for our purposes here we can take any multiple of h, but when we examine quantum gases and take the high temperature limit (where quantum effects are negligible), the choice here provides the correct result.

represents the number of states lying between E and $E + \mathrm{d}E$.

Second method:

We use the delta function representation here, change variables to

$$\epsilon = \frac{p^2}{2m} \Rightarrow p = \sqrt{2m\epsilon} \Rightarrow \mathrm{d}p = \frac{1}{2}\sqrt{\frac{2m}{\epsilon}}\mathrm{d}\epsilon \ . \tag{2.11}$$

Then:

$$\begin{aligned} W(E,V) &= \int \frac{\mathrm{d}^3\mathbf{q}\,\mathrm{d}^3\mathbf{p}}{\mathsf{h}^3}\ \delta\left(E - \frac{p^2}{2m}\right) \\ \text{(Use polars.)} \quad &= \frac{V}{\mathsf{h}^3}\int_0^\infty \mathrm{d}p\ 4\pi p^2 \delta(E-\epsilon) \\ \text{(Use Eq. (2.11).)} \quad &= \frac{4\pi}{2}\frac{V}{\mathsf{h}^3}\int 2m\epsilon\sqrt{\frac{2m}{\epsilon}}\delta(E-\epsilon)\ \mathrm{d}\epsilon \\ \Rightarrow \quad W(E,V) &= \frac{2\pi V}{E}\left(\frac{2mE}{\mathsf{h}^2}\right)^{3/2} \end{aligned} \tag{2.12}$$

in agreement with Eq. (2.10). We may interpret this result more readily if we note that $(\mathsf{h}^2/2m)/E = \lambda(E)$, the de Broglie wavelength for a particle with energy E. Thus

$$W(E,V) = \frac{2\pi}{E}\frac{V}{\lambda^3(E)}\ ,$$

which has dimensions of inverse energy and is the *density of states*. The final factor is the number of wavepackets of the right energy which would fit into V.

For the sphere in phase space for N particles in $6N$ dimensions, we find (Appendix B), where $L = V^{1/3}$ and S_n is the area of the hypersphere of unit radius in n dimensions,

$$\begin{aligned} W(E,V,N) &= \frac{S_{3N}}{2E}\left(\frac{2mL^2\ E}{\mathsf{h}^2}\right)^{3N/2} \\ &\underset{N\to\infty}{\sim} \frac{1}{2E}\sqrt{\frac{3N}{\pi}}\left(\frac{\mathrm{e}2\pi}{3}\frac{2mL^2}{\mathsf{h}^2}\frac{E}{N}\right)^{3N/2} \ . \end{aligned} \tag{2.13}$$

Setting $N = 1$ in Eq. (2.13) yields Eq. (2.12).

2.5 Status of ergodic hypothesis (Nonexaminable)

Experimentally statistical mechanics almost always works, so almost always one might say the ergodic hypothesis is a correct assumption.

When does it *not* work?

- The solar system shows that if one has other conserved quantities – such as angular momentum – the whole surface $E = \mathcal{H}(Q, P)$ is not explored: the plane of the Earth's orbit does not change with time, although the energy would be the same if the plane of the orbit was perpendicular to the plane of the current orbit.

- Consider the potential in one dimension

$$V(q) = -\frac{a}{2}q^2 + \frac{b}{4}q^4 \ .$$

  This has two minima at positions

$$q_\pm = \pm\sqrt{\frac{a}{b}} \ ,$$

  For $E \leq 0$, If the system initially has $q < 0$, it will remain there and similarly for $q > 0$. So the system does not explore all configurations consistent with the energy being fixed, i.e. it is not ergodic.

  However for $E > 0$, the system may explore both positive and negative values of q. So the system is plausibly ergodic for $E > 0$.

Proving that systems are ergodic is very hard see the latter parts of this article. Indeed there is a theorem from mathematics that almost all systems are not ergodic. But this maybe because a randomly selected potential energy function will have local minima at high energies – which stops ergodicity – like the second example above.

There is also the deeper issue: taking a gas as an example, why is it possible to specify the equilibrium state in terms of variables, for example p and V (disregarding the number of moles)?

2.6 Summary

In this section we have always dealt with systems whose energy is conserved.

- Large numbers – eg of atoms – leads to simplifications, such as averages of quantities almost always being observed.

- We showed how to count numbers of configurations for discrete variables.

- The ergodic hypothesis (which almost always works) allows an average over phase space to be constructed for physical quantities, instead of having to integrate the equations of motion for many variables.

- We examined simple cases of the counterpart to counting numbers of configurations for continuous variables.

Appendices

A Quantitative analysis of large number coin tosses

Let us be more quantitative, and approximately find the maximum in this function (plausibly at $n = N/2$) and the width of the peak centred at the maximum. It is convenient to use $\ln W(N,n)$ and Stirling's approximation:

$$\ln N! \underset{N\to\infty}{\sim} N(\ln N - 1) \ , \tag{A.1}$$

derived in the Appendix A of Baierlein in a rough and ready manner.

We now pretend that n is a real number, not an integer, to find the stationary point of W:

$$\begin{aligned} \ln W(N,n) &\underset{N\to\infty}{\sim} N(\ln N - 1) - (N-n)(\ln(N-n) - 1) - n(\ln n - 1) \\ \Rightarrow \qquad \frac{\partial}{\partial n} \ln W(N,n) &\simeq (\ln(N-n) - 1) + 1 - (\ln n - 1) - 1 = 0 \\ \Rightarrow n &= \frac{N}{2} \ , \end{aligned}$$

To determine the width of the distribution around $n = N/2$, it is useful to consider the function

$$f(x) = \mathrm{e}^{-(x/\Delta)^2} \quad \Rightarrow \quad \ln f(x) = -(x/\Delta)^2 \ . \tag{A.2}$$

Δ is the width of the Gaussian.

We may write a Taylor expansion

$$\begin{aligned} \ln W(N,n) &= \ln W\left(N, n = \frac{N}{2}\right) + \underbrace{\frac{\partial}{\partial n} \ln W\left(N, n = \frac{N}{2}\right)}_{=0} \left(n - \frac{N}{2}\right) + \\ &\qquad + \frac{1}{2}\frac{\partial^2}{\partial n^2} \ln W\left(N, n = \frac{N}{2}\right)\left(n - \frac{N}{2}\right)^2 + \cdots \\ &\simeq \ln W\left(N, n = \frac{N}{2}\right) + 0 + \frac{1}{2}\frac{\partial^2}{\partial n^2} \ln W\left(N, n = \frac{N}{2}\right)\left(n - \frac{N}{2}\right)^2 \end{aligned}$$

Then noting that

$$\begin{aligned} \ln W\left(N, n = \frac{N}{2}\right) &\simeq N(\ln N - 1) - \frac{N}{2}\left[\ln\frac{N}{2} - 1\right] - \frac{N}{2}\left[\ln\frac{N}{2} - 1\right] \\ &\simeq N \ln N - N \ln(N/2) \\ &= N \ln 2 \ . \end{aligned}$$

And for the second derivative:

$$\frac{\partial^2}{\partial n^2}\ln W\left(N,n\right) = -\frac{1}{N-n} - \frac{1}{n}$$
$$\Rightarrow \quad \frac{\partial^2}{\partial n^2}\ln W\left(N, n=\frac{N}{2}\right) = -\frac{2}{N} - \frac{2}{N} = -\frac{4}{N} \ .$$

So we may write $W(N,n)$ in the vicinity of $n = N/2$ as

$$W(N,n) \simeq \exp\left[N\ln 2 - \frac{4}{N}\left(n-\frac{N}{2}\right)^2\right] = 2^N \exp\left[-\frac{4}{N}\left(n-\frac{N}{2}\right)^2\right] \ . \tag{A.3}$$

Comparing Eq. (A.2) and Eq. (A.3) we see the width of the peak in $W(N,n)$, at $n = N/2$, is $\sqrt{N}/2$. One might think this implies the peak gets *wider* as N grows – indeed it does; but relative to the allowed values of $0 \leq n \leq N$, it shrinks, consistently with the data from Baierlein, see Table (1).

B Number of states per unit energy for a large number of particles (non-examinable)

The total energy, E, is fixed, and for an ideal gas

$$E = \sum_{n=1}^{N} \frac{\mathbf{p}_n^2}{2m} = \frac{p^2}{2m} \ ,$$

where $p^2 = \sum_{n=1}^{N} \mathbf{p}_n^2$.

Then the total number of points in phase space which have energy E, $W(E,V,N)$ is

$$W(E,V,N) = \int \prod_{n=1}^{N} \frac{\mathrm{d}\mathbf{q}_n \mathrm{d}\mathbf{p}_n}{\mathsf{h}^{3N}} \delta\left(E - \frac{p^2}{2m}\right)$$
$$= \frac{V^N}{\mathsf{h}^{3N}} \int \prod_{n=1}^{N} \mathrm{d}\mathbf{p}_n \delta\left(E - \frac{p^2}{2m}\right) \ .$$

Now use polar coordinates in $3N$ dimensions - as in problem 1.8. There it is found that the hyperarea of the unit radius hypersphere in $3N$ dimensions, S_{3N}, is

$$S_{3N} = \frac{2\pi^{3N/2}}{\Gamma(3N/2)} \quad \text{and} \quad S_{3N} \underset{N\to\infty}{\sim} \sqrt{\frac{3N}{\pi}}\left(\frac{2\pi \mathrm{e}}{3N}\right)^{3N/2} \ .$$

Then:

$$W(E,V,N) = \frac{V^N}{\mathsf{h}^{3N}} S_{3N} \int_0^\infty \mathrm{d}p \ p^{3N-1} \delta\left(E - \frac{p^2}{2m}\right) \ .$$

Now let $\epsilon = p^2/(2m)$, so that:

$$p = \sqrt{2m\epsilon}\ ; p^{3N-1} = (2m\epsilon)^{(3N-1)/2}\ ; \mathrm{d}p = \frac{1}{2}\frac{\sqrt{2m}}{\sqrt{\epsilon}}\ \mathrm{d}\epsilon\ .$$

Thus upon substitution, and using $V = L^3$

$$\begin{aligned} W(E,V,N) &= \frac{1}{2}\frac{V^N}{\mathsf{h}^{3N}} S_{3N} \int_0^\infty \mathrm{d}\epsilon\ (2m\epsilon)^{(3N-1)/2} \frac{\sqrt{2m}}{\sqrt{\epsilon}} \delta(E-\epsilon) \\ &= \frac{1}{2}\frac{V^N}{\mathsf{h}^{3N}} S_{3N} (2m)^{3N/2} E^{(3N-2)/2} \\ &= \frac{1}{2} S_{3N} \underbrace{\frac{V^N (2mE)^{3N/2}}{\mathsf{h}^{3N}}}_{\text{dimensionless}} \frac{1}{E} \end{aligned}$$

(Use results problem 1.8)

$$\underset{N\to\infty}{\sim} \frac{1}{2}\sqrt{\frac{3N}{\pi}} \left(\frac{2\pi \mathrm{e}}{3}\frac{2mL^2}{\mathsf{h}^2}\frac{E}{N}\right)^{3N/2} \frac{1}{E}\ . \tag{B.1}$$

One may check that if one uses the exact (not asymptotic) result for N=1, the answers derived earlier for one particle are recovered. Eq. (B.1) is used in discussing the nature of the partition function integral in sections 3 and 4.